



Presented by:

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Project Roadmap

1. Graphical Interface Calculator
2. UGV Student Mark & CGPA Generator
3. UGV Bus Route Management System
4. Barishal Doctor Appointment System
5. UGV Course Registration System
6. Barishal Launch Ghat Manager
7. UGV Campus Management System (MVC)

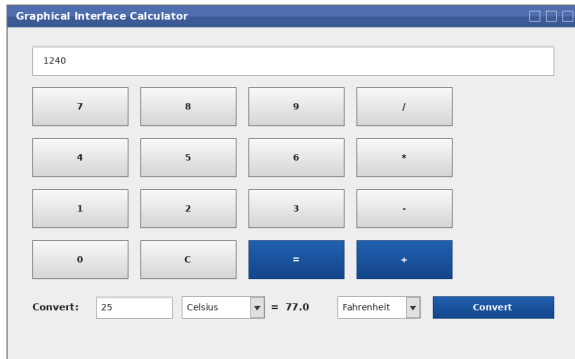
Each project maps to one core OOP topic and ships as a working Java Swing application.

Project 1: Graphical Interface Calculator

OOP Focus: Java syntax, methods, Swing basics

- Buttons, text fields and event handling with `ActionListener`
- Separate methods for add, subtract, multiply, divide
- Built-in unit converter for temperature and weight

No database. Pure syntax and GUI foundations.



Java Swing interface

Project 2: UGV Student Mark & CGPA Generator

OOP Focus: Class, encapsulation, polymorphism

- Private fields with getters and setters hide course data
- Overridden `gradePoint()` for different grading scales
- Live CGPA from a validated Swing form and table

Data held in memory using an `ArrayList`.

The screenshot shows a Java Swing window titled "UGV Student Mark and CGPA Generator". It features a form with four input fields: "Course" (containing "Programming"), "Credit" (containing "3"), "Marks" (containing "85"), and "Scale" (a dropdown menu set to "Standard"). An "Add Course" button is to the right of the "Scale" dropdown. Below the form is a table with the following data:

Course	Credit	Marks	Grade	Point
Programming	3	85	A	4.00
Mathematics	3	78	A-	3.70
Physics	2	72	B+	3.30
English	2	90	A+	4.00

At the bottom of the window, a green box displays the summary: "Total Credits: 10" and "CGPA: 3.78".

Java Swing interface

Project 3: UGV Bus Route Management System

OOP Focus: Generalization and specialization

- Abstract Route class as the general type
- CityRoute, IntercityRoute, ShuttleRoute specialize it
- Each overrides fare and schedule logic

MySQL storage through JDBC starts here.

The screenshot shows a Java Swing window titled "UGV Bus Route Management System". It features a form with the following fields: "Route Type" (a dropdown menu showing "CityRoute"), "Start" (a text box with "Campus Gate"), "End" (a text box with "Nathullabad"), and "Distance (km)" (a text box with "8"). Below these fields are two buttons: "Save Route" and "Show Fare". To the right of the buttons, it says "Calculated Fare: BDT 25". Below the form is a table with the following data:

ID	Type	Start	End	Km	Fare
1	CityRoute	Campus Gate	Nathullabad	8	25
2	IntercityRoute	Barishal	Dhaka	170	550
3	ShuttleRoute	Hostel	Academic Bldg	2	10
4	CityRoute	Rupatoli	Launch Ghat	6	20

Java Swing interface

Project 4: Barishal Doctor Appointment System

OOP Focus: Association, aggregation, composition, 1-to-many, many-to-many

- Hospital owns departments (composition)
- One doctor to many appointments (one-to-many)
- Many doctors to many patients (many-to-many)

Related MySQL tables joined in Swing tables.

The screenshot shows a Java Swing window titled "Barishal Doctor Appointment Management System". It features a form with four dropdown menus: "Department" (set to "Medicine"), "Doctor" (set to "Dr. Rahman"), "Patient" (set to "Karim"), and "Date" (set to "2026-07-10"). Below the form are two buttons: "Book Appointment" and "Cancel". At the bottom, there is a table displaying appointment records.

ID	Doctor	Patient	Department	Date	Status
11	Dr. Rahman	Karim	Medicine	2026-07-10	Booked
12	Dr. Sultana	Nabila	Cardiology	2026-07-11	Booked
13	Dr. Rahman	Hasan	Medicine	2026-07-12	Pending
14	Dr. Alam	Riya	Pediatrics	2026-07-12	Booked

Java Swing interface

Project 5: UGV Course Registration System

OOP Focus: Constructor, this, super

- Overloaded constructors: empty, partial and full
- this resolves field and parameter name clashes
- super() calls the parent Person constructor

Objects built from the form based on filled fields.

The screenshot shows a Java Swing window titled "UGV Course Registration System". It contains several input fields and a list of courses. The "Student Name" field is filled with "Md. Sakib", and the "Student ID" field is filled with "CSE-2101". The "Program" dropdown menu is set to "B.Sc in CSE", and the "Semester" dropdown menu is set to "Spring 2026". A blue "Register Student" button is located below the input fields. To the right, under the heading "Select Courses", there is a list of five courses with checkboxes: "Object Oriented Prog" (checked), "Data Structures" (checked), "Database Systems" (unchecked), "Web Engineering" (checked), and "Algorithms" (unchecked). Below the "Register Student" button, there is a light blue box titled "Registration Summary" containing the text "Md. Sakib (CSE-2101)" and "B.Sc in CSE, Spring 2026, 3 courses".

Java Swing interface

Project 6: Barishal Launch Ghat Manager

OOP Focus: Inheritance (all types)

- Single, multilevel and hierarchical inheritance
- Multiple inheritance of type through interfaces
- `describe()` overridden across the hierarchy

Launch, speedboat and trawler records in MySQL.

ID	Type	Name	Capacity	GPS
1	Launch	MV Sundarban	700	Yes
2	LuxuryLaunch	MV Kirtonkhola	900	Yes
3	Speedboat	Fast-01	12	No
4	Trawler	Jejepara-3	40	No

describe():
Launch MV Sundarban, capacity 700, GPS enabled, passenger rated.

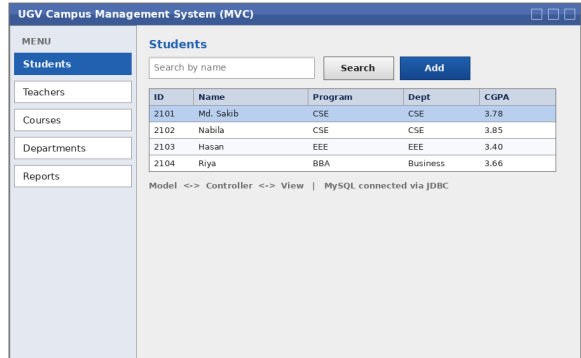
Java Swing interface

Project 7: UGV Campus Management System (MVC)

OOP Focus: All topics combined with MVC

- Model holds data and JDBC logic
- View is the Swing screen set
- Controller wires events between them

Final capstone integrating every concept.



Java Swing interface

- Projects 1 to 2 build syntax, classes and encapsulation with no database.
- Projects 3 to 6 add database work through JDBC and MySQL, one OOP topic each.
- Project 7 combines everything into one MVC application.

Thank You

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